

CODE SPORT



Welcome to CodeSport! In this month's column, we feature a set of programming questions, as requested by a few of our student readers.

In last month's column, we discussed a few of the programming errors that can lead to software bugs. In this context, one of our readers, Balamurugan, had emailed us a pointer to his favourite open source static analysis tool for Java, called FindBugs, which is available at <http://findbugs.sourceforge.net/>. There is also a good discussion on programming errors that lead to hard-to-detect bugs, by William Pugh, the author of 'FindBugs', available at <http://www.cs.umd.edu/~pugh/MistakesThatMatter.pdf>. I would encourage our readers to go through this presentation.

Another reader, Raghunath, had sent a pointer to Clang (<http://clang-analyzer.lvm.org/>), which can be used to detect coding errors in C programs. Clang is not a static analysis tool *per se*, but is a compiler front end for C, and uses LLVM (www.llvm.org) as the back end. The Clang front-end provides static analysis tool capabilities to detect coding errors. A standalone version of the tool is also available. We thank our readers Balamurugan and Raghunath for sharing these pointers to their favourite tools for detecting software bugs. If you have a favourite static analysis/runtime tool to detect programming errors, please do send me a note. I will share your inputs with other readers in next month's column.

A few of our student readers had written in with a request to feature a set of programming interview questions that they could use for practice. Therefore, in this month's column, we take a break from our discussion of software bugs, and instead feature a set of programming problems. We will continue our discussion on static and runtime tools to detect software bugs in next month's column.

Before we get started on this month's questions, I wanted to point our readers to a useful link related to preparing for an interview that I came across. We all know that many of our readers dream of getting to

join companies like Google or Facebook. In fact, MIT now offers a course on how to crack the programming interviews in companies like Google or Facebook. The link to the course is available at <http://courses.csail.mit.edu/iap/interview/materials.php>.

This month's programming problems

- Given two sorted integer arrays X and Y of size m and n , you are asked to find the k th smallest element in the new array Z formed by merging X and Y .
 - Can you give a solution which is $O(m+n)$?
 - Can you give a solution which is $O(k)$?
- Given two sorted integer arrays X and Y of size m and n , you are asked to find the intersection of the two arrays.
 - What is the complexity of your solution?
 - Can you give a solution which is $O(m+n)$, and using constant extra space?
- We are familiar with the queue data structure, which supports *push_back* and *pop_front* operations in constant time. Can you design a queue data structure that supports the operation *extract_min* in constant time, along with *push_back* and *pop_front* operations?
- Given a binary tree:
 - Write a program to determine whether it is a binary search tree.
 - If the binary tree is not a binary search tree, can you find a binary search tree with the largest number of nodes in the binary tree? The largest BST may or may not include all of its descendants; i.e., the largest BST you find need not be a sub-tree of the binary tree.
- You are representing a set of integers using a linked list. You are asked to support the array index operator on the linked list. Given an index i , you need to support the following operations: